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Implement a pre-trained CNN model as a feature extractor using transfer learning model

Aim: To implement a pre-trained CNN model as a feature extractor using transfer learning model.

Objective:

1. To understand how a CNN pre-trained on ImageNet can extract visual features.

2. To freeze convolutional layers and use their output features for new tasks.

Pseudocode:

1. Import libraries -
2. Load a pre-trained CNN
3. Freeze all convolutional layers to prevent retraining
4. Replace final classifier layer for new number.
5. Load small dataset (CIFAR-10)
6. Extract features
7. Train only new classifier head
8. Evaluate accuracy

Architecture

target 'image' data
cell labels

conv2d - 1

maxpool2d - 1

maxpool2d - 3

fc - 1 (1024)

fc - 1 (500, 75)

classifiers

output

Observation:

+ The pre-trained ResNet15 model was successfully loaded.

+ Frozen layers acted as fixed features extract, capturing low-level edge & corners - features.

+ Training was significantly faster because only the final layer's parameter were updated leading to stable convergence.



Result:

✓ Successfully implemented pre-trained CNN model

Output

epoch 1 Loss : 0.8348

epoch 2 Loss : 0.6229

epoch 3 Loss : 0.5937

epoch 4 Loss : 0.5797

epoch 5 , Loss = 0.5699